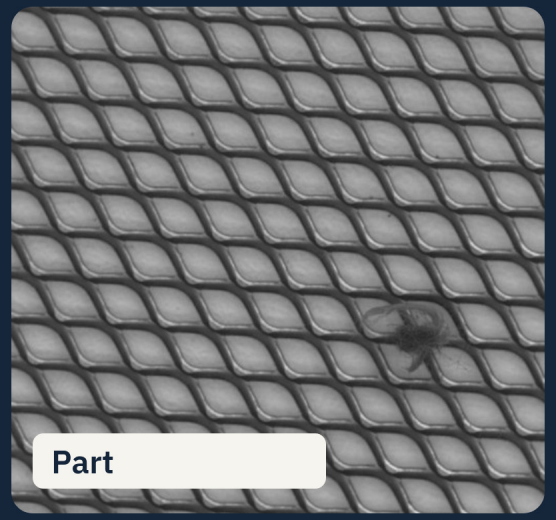
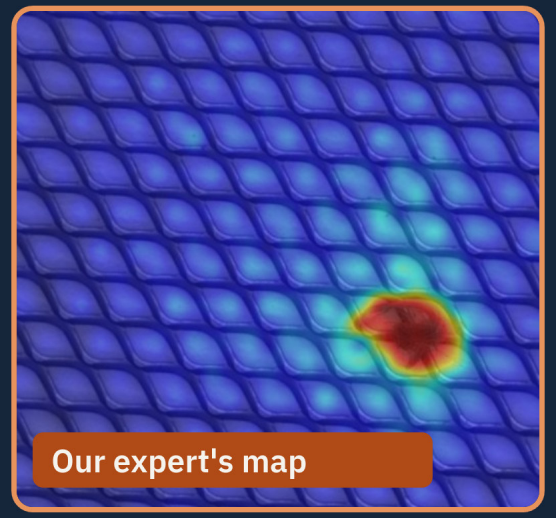
IA025 · DEEP LEARNING · UNICAMP 2026

# OneRef

Preliminary results: the pipeline runs end to end, and the visual expert is measured on real MMAD data




**Team:** Peter Roque Vilchez · Charles Maxime Doudy

Delivery II · November 12, 2026

CONTEXT

# One good photo, one part, a question



→ 1 · Visual expert → 2 · Two readers → 3 · Verifier → 4 · Vote

**Benchmark**

MMAD: 8,366 images, 39,670 multiple-choice questions, 7 subtasks, strict 1-shot

**Baseline to beat**

Qwen3-VL-8B alone, 73.9%, and a same-cost majority vote

**Rules**

No training, models ≤ 8B, tune on dev only, test run once

Hypothesis: the multi-agent system beats both baselines with significance (McNemar test). A well-measured negative result is also valid.

DEVELOPMENT · WHAT WE BUILT

# The full workflow exists as code

**Visual expert**
DINOv3, patch distance
**Ran on all 835 dev**

**Readers A · B**
Qwen3-VL · InternVL3 via vLLM
**Ran on real data**

**Verifier**
rules + LLM critic, reopen ≤ 1
**Ran on real data**

**Aggregator**
weighted vote per subtask
**Ran on real data**

**Orchestrator**
one model per pass, cache, resume
**Ran on real data**

## 21 / 21

GPU-free tests pass (schemas, critic, vote, metrics, resume)

## 835 / 7,531

dev / test images, stratified, seed fixed and committed

## 2.1 GB

to fetch dev instead of 28 GB: only the needed files, read out of the remote zips

Messages between agents are JSON validated with Pydantic; invalid answers are retried once, then count as errors. Code: github.com/RVilchezzz/OneRef

DEVELOPMENT · FIRST REAL NUMBERS

# The expert works on aligned photos only

| Source (dev) | Images | AUROC v2 | AUROC v3 | In mask v2 | In mask v3 |
|---|---|---|---|---|---|
| MVTec AD | 166 | 0.88 | 0.84 | 51% | 77% |
| VisA | 215 | 0.77 | 0.78 | 42% | 41% |
| MVTec LOCO | 156 | 0.64 | 0.62 | 37% | 43% |
| GoodsAD | 296 | 0.55 | 0.51 | 12% | 10% |
| All | 835 | 0.61 | 0.59 | 33% | 40% |

**AUROC:** chance that a defective part scores higher than a normal one (0.5 = chance). **In mask:** the map's hottest point falls on the true defect.

**DINOv3 localizes better**

77% vs 51% on MVTec AD: it knows where to look, which is the expert's job

**GoodsAD ≈ chance**

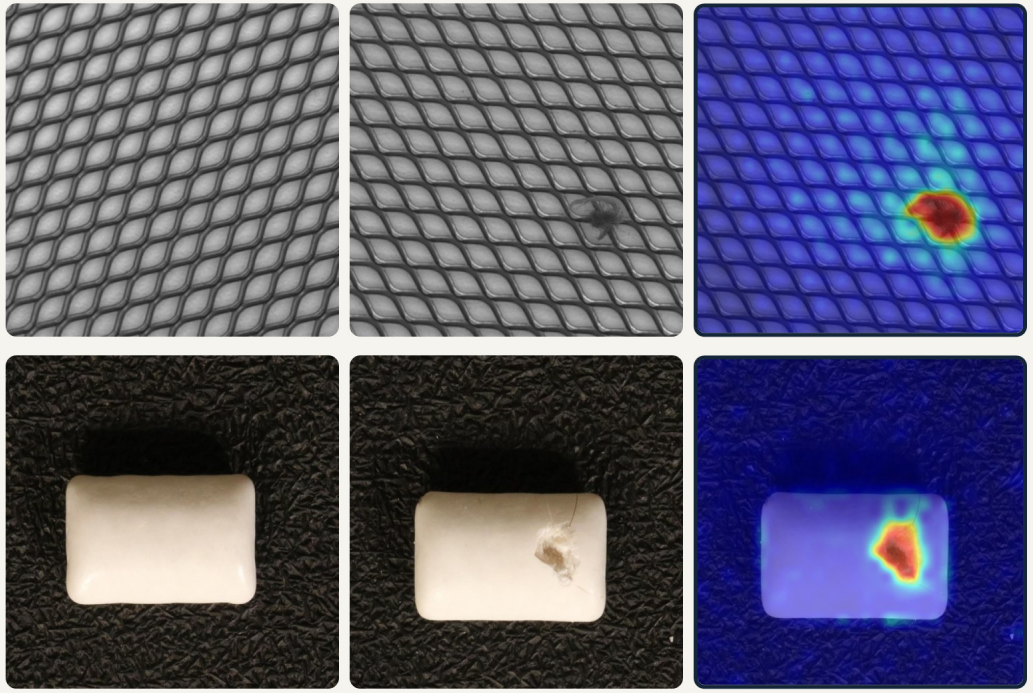
the one reference is often another product, not the same one

Frozen DINOv2-base and DINOv3 ViT-B/16, query vs random_templates[0], dev split only, one run. RTX 3060 Laptop: 0.35 s per image. Masks used only to score.

DEVELOPMENT · WHAT WORKED, WHAT DID NOT

# Reference · part · our DINOv3 map

✓ Worked: defect found



× Failed: reference is another product




Real dev images: MVTec AD grid, VisA chewing gum, GoodsAD food box, MVTec LOCO juice. Images: MMAD (CC BY-NC-SA 4.0, CC BY 4.0, research use).

DEVELOPMENT · WHAT WENT WRONG

# Four problems found, and what we do about them

| Problem | Evidence | Fix / plan |
|---|---|---|
| Loader never tried on real data | crashed on “Anomaly Detection”, the name mmad.json uses | fixed + test with the 9 real names |
| Expert threshold off-scale for DINOv3 | distances 0.11–0.31 vs range 0.2–0.8: flag never fires (bal. acc. 0.49) | recalibrated on dev: 0.49 → 0.61 (in-sample) |
| Single reference ≠ same product | GoodsAD AUROC 0.51, peak on defect 10% | trust the expert less there: weights fixed on dev |
| 8B readers not run yet | Colab not executed; local 4B run: 18 images only | 50-question cost test, then full dev |

Lesson: unit tests with fake data passed 20/20 while the real file broke the first line. Run on real data early.

DEVELOPMENT · FIRST END-TO-END RUN

# All five agents ran on real MMAD images

Expert 18 s → Reader A 323 s → Reader B 322 s → Verifier: 4 reopened → Vote

| Subtask (88 questions) | Accuracy |
|---|---|
| Anomaly discrimination (balanced) | 52.5% |
| Defect classification · description | 62.5% · 62.5% |
| Defect localization | 50.0% |
| Defect analysis | 87.5% |
| Object classification · analysis | 100% · 83.3% |
| Mean of 7 | 71.2% |

**0 errors in 184 calls**

all answers valid JSON; ~3.7 s per question

**Readers agree 87/88**

same model twice: why we need two different families

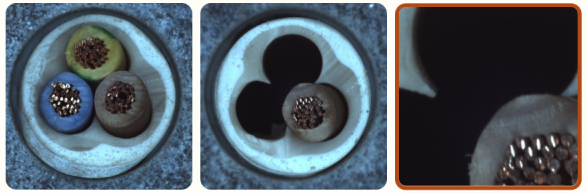
**95% CI [54%, 86%]**

18 images: a working pipeline, not a result

Smoke test on the laptop (RTX 3060, 6 GB): 18 dev images, 2–4 per source; both readers = Qwen3-VL-4B quantized (Ollama); rules-only verifier. The 8B readers run on Colab.

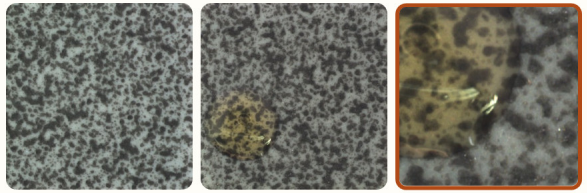
DEVELOPMENT · EXAMPLES FROM THE RUN

# Four real cases, four behaviours



**✓ Cable: 5/5 right**

“Missing component, in the central part”



**↻ Tile: verifier reopened**

readers said bottom-left, expert bottom-center; still wrong (center-left)




**× Can: false alarm**

good part called defective: the reference is another brand



**× Pushpins: followed the hint**

“the expert hint indicates the top right”: wrong place

**Takeaway:** the expert's hint is powerful both ways, so its weight must depend on how reliable it was on dev.

Each card: reference · part · zoom the expert sent to the readers (orange). From the 18-image run: MVTec AD cable and tile, GoodsAD can, MVTec LOCO pushpins.

PLANNED VS. DONE

# What the plan promised for Nov 12

| Planned (Delivery I) | Status | Detail |
|---|---|---|
| MMAD download | Done | dev subset, 2.1 GB, 2,084 files |
| Dev and test splits fixed | Done | 835 / 7,531 images, committed |
| DINOv3 expert with anomaly maps | Done | all 835 dev images, per-source scores |
| Evaluation harness | Partial | metrics, McNemar, bootstrap coded; check vs official summary.py pending |
| B1 and B2 on dev | Pending | readers coded and tested, not run |
| Colab throughput measured | Pending | first task this week |

**3 / 6**

items done

**+ extra**

full multi-agent workflow coded and already run end to end on 18 dev images (planned for Nov 12–19)

NEXT STEPS

# Same goal, adjusted path to Nov 26

**NOV 12–15**

### Readers on dev

50-question cost test on Colab L4, then B1 and B2 on the 3,937 dev questions

**NOV 15–19**

### Calibrate on dev

expert threshold, vote weights per reader and subtask, prompts; official summary.py check

**NOV 19 · FREEZE**

### Test, run once

S vs B1–B4, ablations; ~1,000–2,000 test images depending on measured cost

**NOV 26**

### Final

McNemar + bootstrap, error analysis per source, report

**Adjustment:** the expert's vote will depend on how reliable it was on dev for each subtask, and results are always reported per source (GoodsAD and LOCO apart).

ONEREF · DELIVERY II

# Thank you! Questions?

Code and results: github.com/RVilchezzz/OneRef

Peter Roque Vilchez · Charles Maxime Doudy

